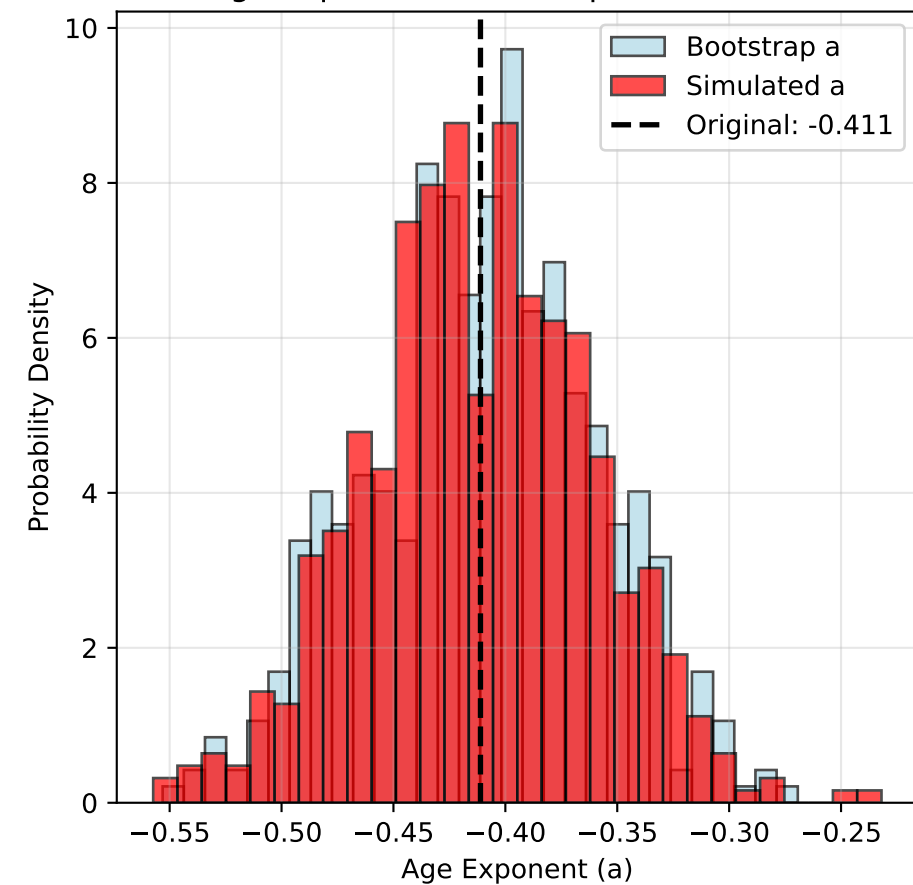
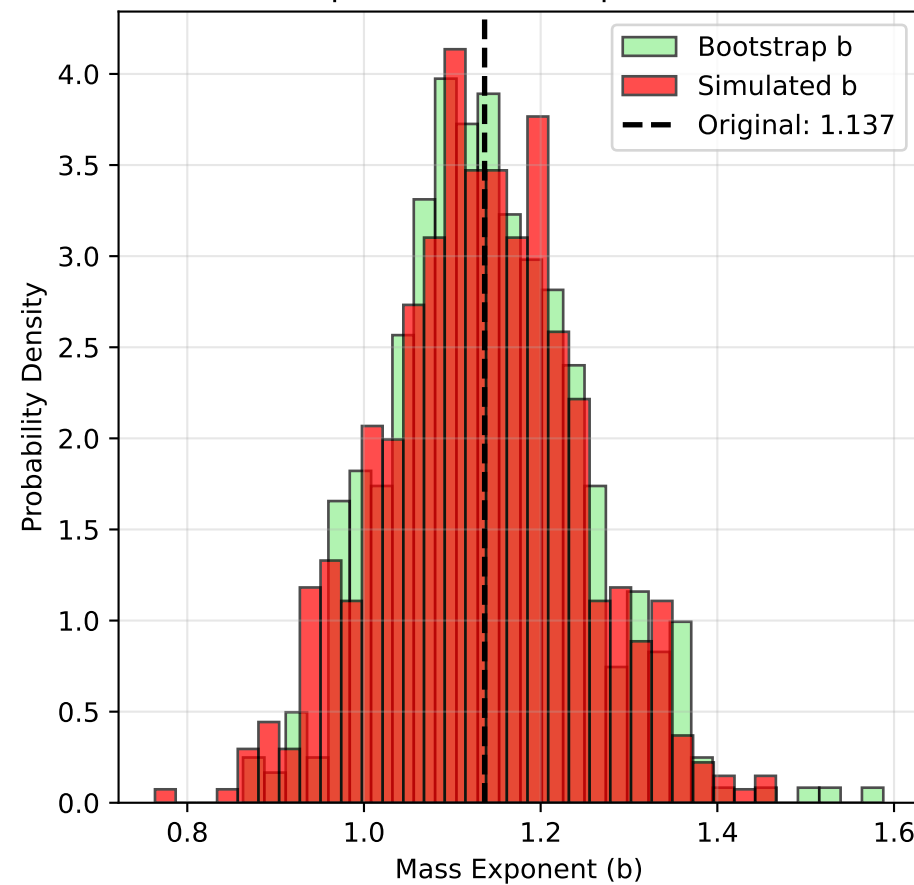


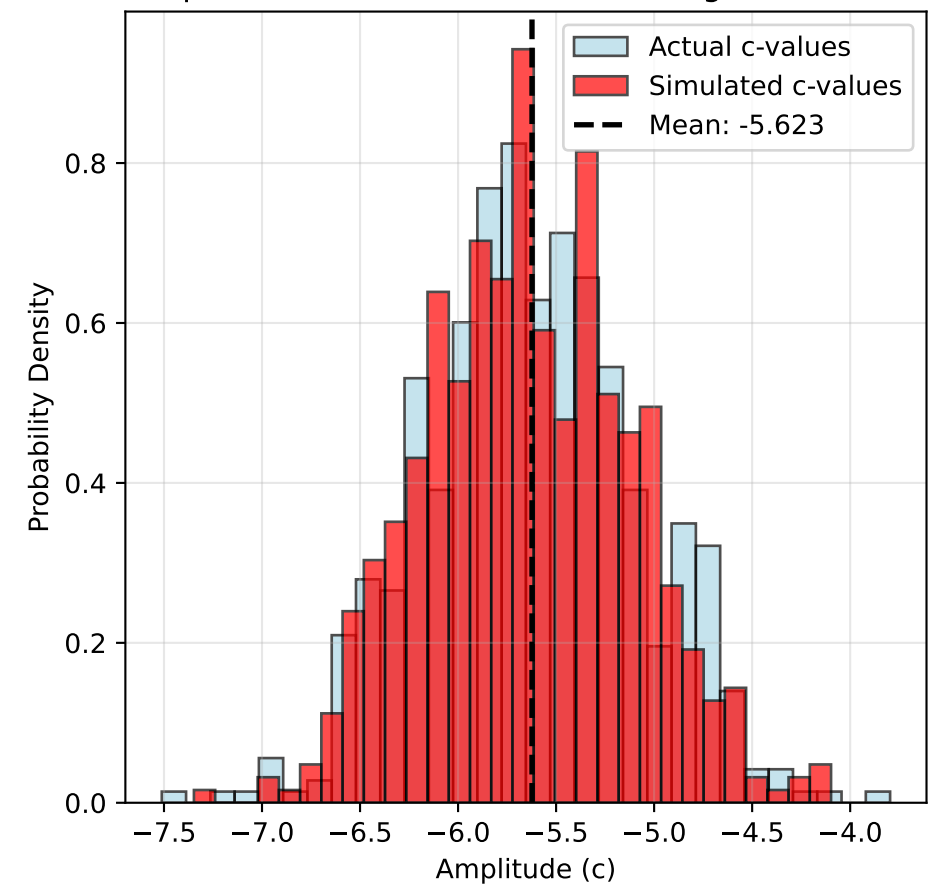
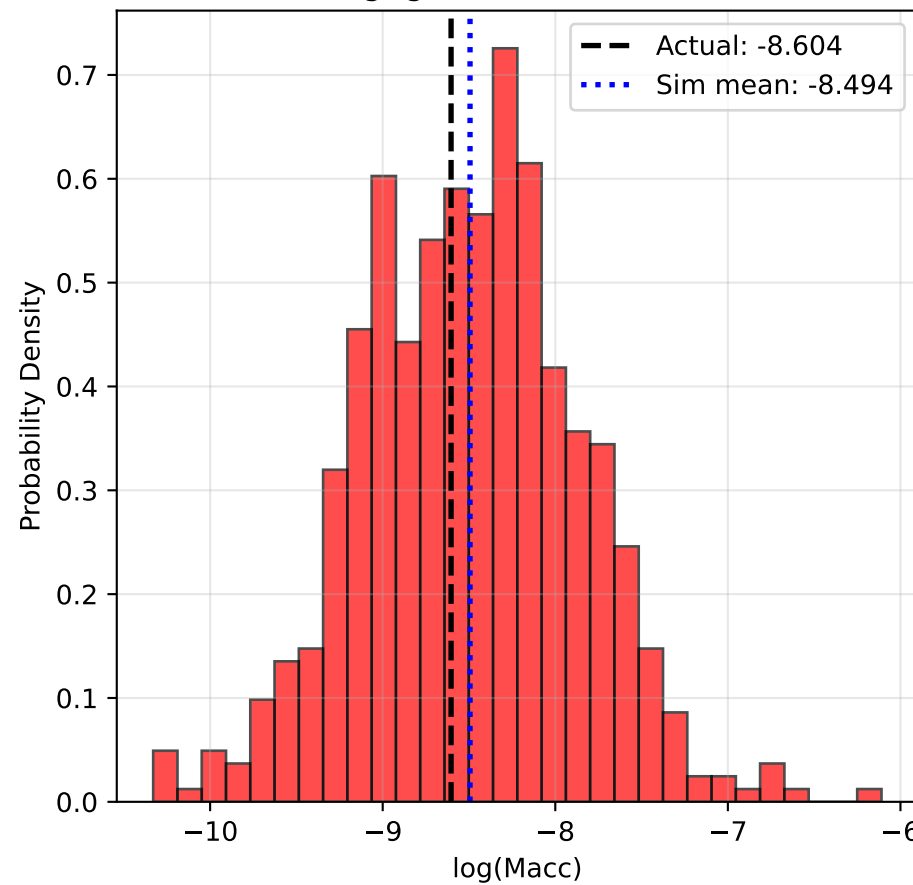
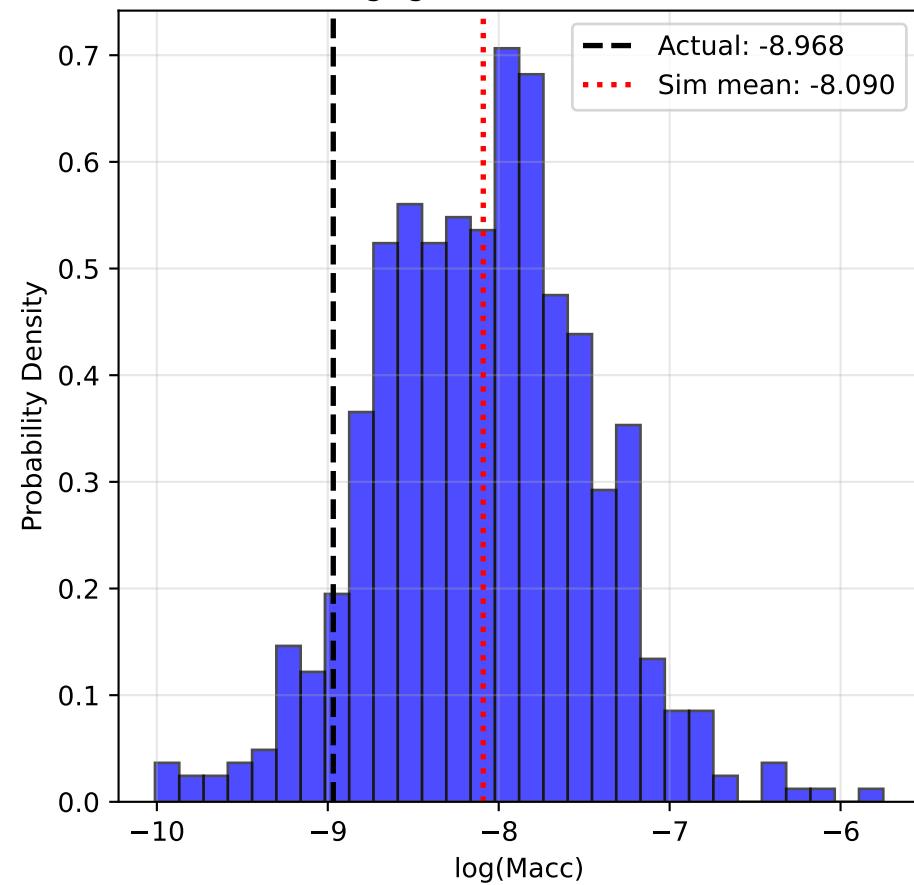
Age Exponent: Bootstrap vs Simulated



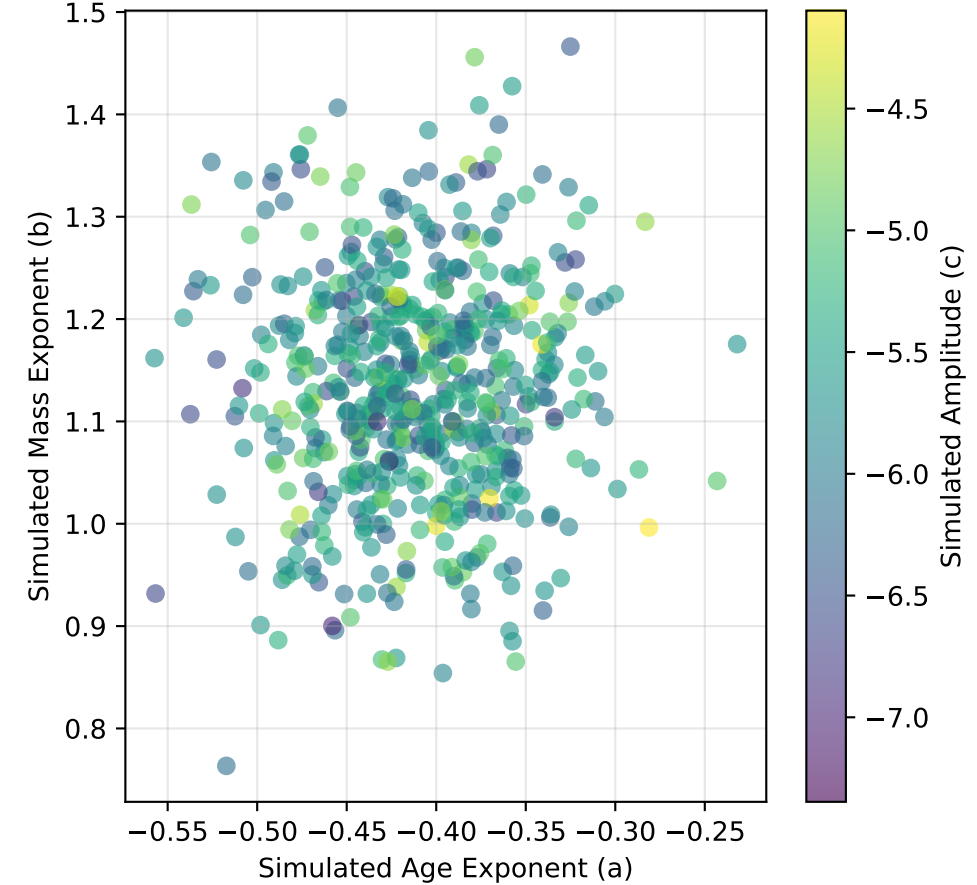
Mass Exponent: Bootstrap vs Simulated



Amplitude: Actual vs Simulated (Original Method)

Star 1: logAge=6.10, Mstar=0.50 $M_{\odot}$ Star 2: logAge=6.16, Mstar=1.20 $M_{\odot}$ 

Simulated Parameter Correlations



PARAMETER UNCERTAINTY ANALYSIS REPORT

Combining Bootstrap and Residual Methods

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METHODOLOGY SUMMARY:

- a,b parameters: Bootstrap resampling (statistical uncertainty)
- c parameter: Original residual method (physical scatter)

FINAL UNCERTAINTY ESTIMATES:

Age exponent (a): -0.410 ± 0.051

Mass exponent (b): 1.140 ± 0.109

Amplitude (c): -5.623 ± 0.541

MODEL ROBUSTNESS:

Bootstrap iterations: 500

Parameter consistency (KS p-values):

a: 0.369 ✓

b: 0.589 ✓

c: 0.750 ✓

KEY INSIGHTS:

- Total Macc uncertainty: ± 0.555 dex
- Parameter uncertainty dominates when predicting individual stars
- Intrinsic scatter represents real astrophysical diversity